START FROM SCRATCH / HYPER-PACED LEARNING

Week 8: Connect the whole stack

Explain how governance, model access, and the live agent/data flow fit together - and how a request remains controlled from entry to outcome.

WHAT YOU ALREADY HAVE

A built, observed, tested SRE Agent

The cohort has moved from AI fluency to model calls, RAG, tool use, multi-agent patterns, observability, and security testing. Week 8 connects these capabilities into one system view.

THIS WEEK'S PLAN

Map three architecture lenses

Use the Control Plane (GRC), Model Gateway, and Data Plane to map ownership and policy, model traffic, and the runtime work that handles requests, retrieval, tools, and results.

RED FLAGS / TESTING

Trace controls as well as requests

Look for bypass paths, direct provider credentials, unclear policy ownership, untracked data movement, missing identity context, weak audit evidence, and gaps between a policy decision and runtime enforcement.

EXPECTED OUTCOME

One explainable end-to-end system

Present an annotated architecture and walk through a request from user to agent, gateway, model, data/tools, response, and audit/evaluation feedback. Explain who governs each boundary.

The model is only one component. A credible AI system also needs governed access, protected data, observable execution, and accountable decisions.

Team rhythm: prepare independently, align with your team and lead, build together, then share. Each team has 15 minutes: 5 minutes to demo and 10 minutes for discussion and feedback.

PREPARE BEFORE YOU BUILD

Put architecture and governance in context

Use the program's three-lens model to understand responsibilities, not as a claim that every vendor or platform implements identical boundaries.

01

AI Risk Management Framework: Generative AI Profile

NIST

Use the Govern, Map, Measure, and Manage framing to think about risk ownership, intended context, evaluation, and response across the AI lifecycle.

Open the NIST profile

02

Use a gateway in front of model deployments

Microsoft Azure Architecture Center

Read as one vendor-specific example of centralized model routing, security segmentation, reliability, observability, and quota governance. The concepts are useful even if your team's prototype uses another provider.

Review the gateway pattern

Bring these questions to your team

- Which decisions belong to organizational governance, and which are enforced on each runtime request?
- Where does a model gateway sit in the request path, and which controls should it centralize?
- What data does the agent read or write, which identity is used, and how can the flow be audited end to end?

The Control Plane, Model Gateway, and Data Plane are the program's teaching lenses for this capstone. Real architectures may distribute components differently; show the actual responsibilities and request paths in your design.

THE WHOLE SYSTEM

Three lenses, one governed request

Keep the responsibilities distinct. Governance sets expectations and accountability; the gateway mediates model traffic; the data plane executes the application and agent work.

Control Plane / GRC	Model Gateway	Data Plane
Defines ownership, acceptable use, risk decisions, policy, identity standards, approvals, evidence, and review cadence.	Mediates model access: authentication, routing, allowed model choices, limits, policy checks, usage visibility, and provider abstraction.	Handles the live workload: user request, agent orchestration, RAG, tools, operational data, model responses, and the result returned to the user.

Walk one request through the architecture

Stage	Question to answer	Evidence to show
Govern	Who approved the use case, data, and permitted autonomy?	Owner, risk record, policy, and approval path
Enter and identify	Who is calling, and what may that identity access?	Authentication context and authorization decision
Mediate model call	Which model is allowed, and what limits or checks apply?	Gateway route, policy result, quota or usage signal
Execute workload	What does the agent retrieve, decide, and do?	Trace across model, RAG, specialist, and tool steps
Return and learn	How is the outcome evaluated, audited, and improved?	User result, audit event, evaluation, and owner

Keep the model honest about the architecture

Do not imply that GRC, a gateway, or a diagram makes a system safe by itself. Show which component enforces each control, where evidence is recorded, what can bypass it, and how a person responds when the control fails.

COME READY TO DESIGN

Your End-to-End Architecture Canvas

Map the real design, label trust boundaries, and follow one request through all three lenses. Keep the diagram readable to both practitioners and leadership.

1. Use case and owner Who uses the SRE Agent, for what purpose, and who is accountable?	**2. Control Plane / GRC** Which risk, policy, approval, and audit decisions govern it?
3. Model Gateway How are identity, routing, limits, and provider access mediated?	**4. Data Plane** Which agent, RAG, data sources, tools, and services handle the request?
5. Trust boundaries Where does data or authority cross a boundary?	**6. Observability** Which trace, metrics, and evaluation link the lifecycle?
7. Failure and bypass What happens if a control, gateway, model, or tool is unavailable?	**8. Demo** Can you trace a request and identify an owner at every step?

Show the learning

- Trace one SRE request through governance, identity, gateway, model, retrieval/tools, and response.
- Point out where policy is defined, where it is enforced, and where evidence is retained.
- Show one failure or denied request and explain who responds and how the system remains safe.

Connect the architecture to the people who operate it. Keep slides minimal, show the real flow, and make ownership and evidence clear.